

Forecasting the Future

A MODERN ECONOMICS CHALLENGE

HARDING, JOE (UG)

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1.0.0 Binary forecasts:

- I. By 2026, the U.S. will expand its industrial subsidy programs (beyond semiconductor chips) to include at least \$50 billion for domestic production of electric-vehicle batteries and critical minerals (65% probability).
- II. By 2027, China will impose new export controls or higher taxes on strategic minerals (e.g. rare earths, lithium) used in high-tech and clean-energy industries (60% probability).
- III. By 2028, the European Union will enforce an anti-subsidy tariff or impose quotas on imports of Chinese green technology exports to protect its domestic green-tech industry (70% probability).
- IV. By 2029, Japan will have invested over \$65 billion in policies supporting its domestic semiconductor and AI sectors (80% probability).
- V. By 2030, South Korea and Indonesia will both have formally joined the CPTPP trade agreement (65% probability).
- VI. By 2027, India will expand its “Make in India” subsidy program to cover pharmaceuticals and active chemical ingredients, boosting domestic production by at least 25% (55% probability).
- VII. By 2026, U.S. tariffs on Chinese imports of electric vehicles and batteries will remain above the global tariff average (75% probability).
- VIII. By 2028, the United States will negotiate new bilateral trade initiatives within the Americas (expanding USMCA provisions) to create “near-shoring” manufacturing corridors (50% probability).
- IX. By 2032, at least two African nations will sign trade agreements or join blocs (e.g. RCEP or CPTPP) led by Asia’s major economies (30% probability).
- X. By 2029, global merchandise trade as a percentage of world GDP will fall below 40% (45% probability).
- XI. By 2035, U.S. foreign direct investment in China will decline to less than 5% of total outbound FDI (55% probability).
- XII. By 2032, one or more major economies will formally peg or intervene in their currency to keep export prices competitive (35% probability).
- XIII. By 2028, at least half of G20 countries will have formal industrial strategies targeting self-sufficiency in semiconductor manufacturing (50% probability).
- XIV. By 2026, U.S. allies under the Indo-Pacific Economic Framework will agree to coordinated friend-shoring principals stopping key supply-chain links to China (45% probability).
- XV. By 2035, multiple countries will implement rules favouring national producers in clean-tech sectors (70% probability).

- XVI. By 2035, a chunk of FTSE 500 manufacturers will report significant relocation of production from Asia to domestic or regional sites (50% probability).
- XVII. By 2027, the World Trade Organization will have lost its resolution power, reflecting a long-term weakening of multilateral trade governance (75% probability)
- XVIII. By 2029, U.S. real manufacturing output will be at least 10% higher than present levels, driven by policy support for domestic industry (65% probability)
- XIX. By 2035, Russia and China will conduct at least 45% of their bilateral trade in their own currencies (55% probability).
- XX. By 2026, more than 25% of global trade volume in at least one key sector (e.g. steel, autos, or semiconductors) will be subject to tariffs or quotas (70% probability).

2.0.0 Developing a Framework of Modern Mercantilism:

The conventional principles that have dictated the last century of international trade-specialisation according to comparative advantage, liberal and open free markets governed by multilateral institutions, namely the WTO- have been recently eroding. I predict that the upcoming decade will replace these conventions with a new 'Modern Mercantilism'. This section presents an interpretation of the world economy in the decade ahead whereby core drivers of change reshape trade dynamics into a landscape of state-led interventions, shifting geopolitical rivalries and alliances, and a wider push for self/bloc contained sufficiency.

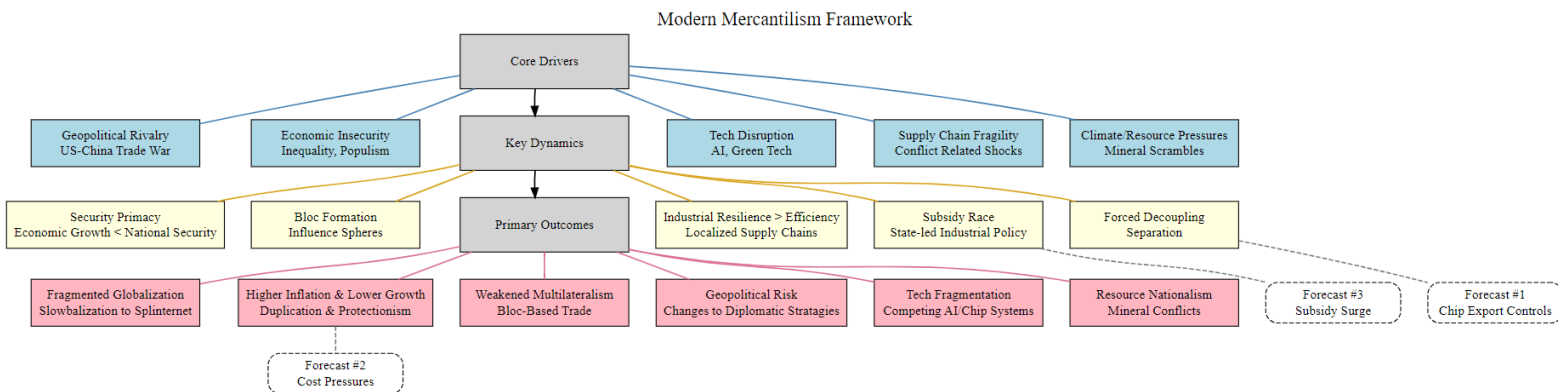


Figure 1

2.1.0 The Shifting Core: From Globalisation to Protectionism

The new trade order, defined by industrial policy driven by subsidies, trade blocs and geopolitical risk management breaks from the neoliberal model of globalisation. It is replaced by a fragmented system where economic security and strategic autonomy are increasingly determining trade flows, such require protectionist policies (Junio, 2025). Protectionist policies-tariffs, subsidies, quotas etc- are a means of supporting infant or legacy industry so that they can realise economies of scale or maintain onshore production common in key industries like energy technology, chip manufacturing or defence manufacturing. All of which are growing in importance in an increasingly fractured world where conflict and fall-out are on the upward trend

2.2.0 East Vs West: The Growing Divide

“With the world now divided among three broad blocs of countries – China-leaning, US-leaning, and nonaligned – both a ‘significant reversal of the gains from economic integration’ and ‘a broad retreat from global rules of engagement’ are on the horizon.”

— IMF Deputy Managing Director Gita Gopinath

As the US, European Union, China and Russia navigate an increasingly polarised world, global trade will undoubtedly begin to mirror these divides. The traditional liberal trade order, which was built on self-determined trading relationships, fractures as nations prioritise secure and strategic interactions over economic growth. At the core of this international factionalism is an unworkable

political dichotomy. In the political domain, Western governments currently work towards preventing further Chinese penetration into their own market: Section 301 tariffs, tightening of CFIUS and the U.K.'s National Security and Investment Act all work towards securing a country's own economy.

UNGA Voting Alignment among Major Economies

Eastern bloc in red, Western bloc in blue | Agreement >50% shown

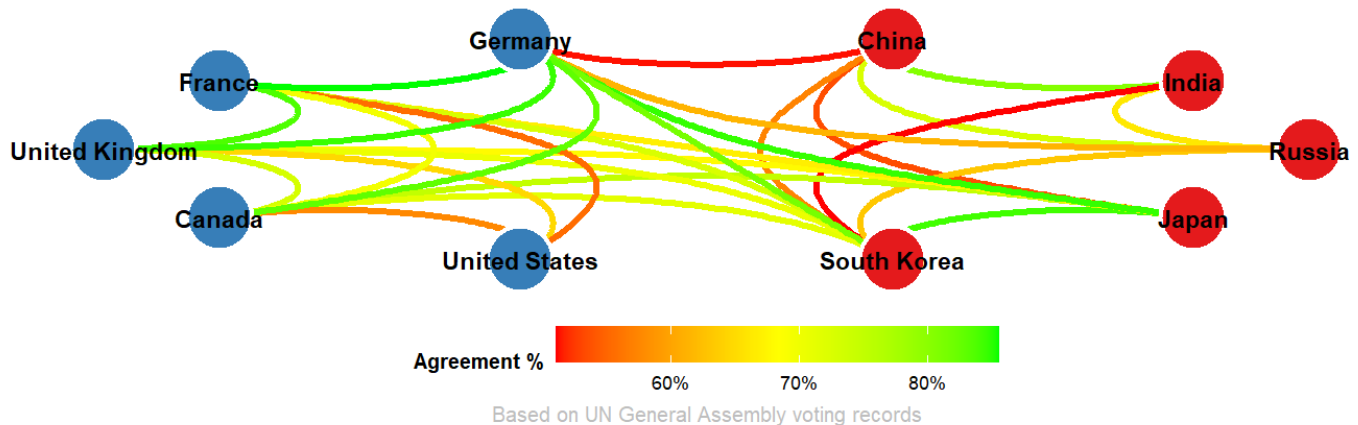


Figure 2

The divide is based on competing political and economic regimes. Western democracies focus on the mutual accountability cycle—elections, independent judiciaries, parliamentary oversight, and a free media—as optimal circumstances for innovation and corruption control. In contrast, the “Beijing Consensus” believes that a meritocratic, one-party system with multi-tiered social-credit incentives and strategic government planning delivers reproducible GDP growth rates, large-scale infrastructure roll-outs, and crisis-response speeds that multiparty democracies cannot replicate. These competing models manifest in competing multilateral areas: the U.S. and its allies champion the Indo-Pacific Economic Framework (IPEF) and the Blue Dot Network, which enforce high-standards on labour and environmental safeguards; China counters with the Regional Comprehensive Economic Partnership (RCEP) and its own high-speed rail corridors, which emphasize non-interference and state-led project delivery.

U.S.-China relations will continue to define the trajectory of global trade, with both likely to expand their tariff restrictions as a mutual desire to decouple, dominate discussion in political and economic forums, this will bring about a new trade order.

2.3.0 A New Trade Order

The shift towards localised trade and bilateral agreements, as countries seemingly wield finance and trade as tools of political influence, will come at the expense of multilateral organisations. The WTO will continue to lose its arbitrating power as the Doha Round of negotiations continues to fizzle out, if another round of negotiations begins it will ultimately follow a similar trend. Bloc Based trade will take precedence, the CPTPP will likely expand, including South Korea and Indonesia. The addition of these manufacturing epicenters is likely to signal a growing asia-centric trade network.

An East Vs West narrative, can narrow into a China Vs U.S. battle where peripheral economies increase in importance as each power postures to expand their trading relations. Regions like

Africa and South America will rise in prominence and will no longer be sidelined and treated as secondary markets for eastern and western economies, but will become a key battleground as the U.S. and China jockey for dominance. India, with its increasing industrial base and advantageous geographical position, will likely see its global trade share grow substantially. By the end of our forecasting period, India will be poised to become a key partner for Eastern and Western powers, with trading routes flowing through the Indian Ocean elevating the country into a major trading hub.

Africa, likewise, will see an influx of foreign investment into their industrial sector. In particular, China will wish to secure a relationship with key mineral exporters by way of expanding its Belt and Road and Digital Silk Road projects. Expect America to further contest these areas and attempt to block Chinese monopsony over these precious minerals, especially if America continues to pursue 'on-shore' microchip and EV industry policy, perhaps by way of an expansion of the G7 B3W projects. That is an initiative to provide an alternative financing option for developing nations that do not rely on Chinese funds. Similarly, we can expect the U.S. to undertake 'friendshoring' policies where trade links are determined by political allegiance. This will entail a duplication of manufacturing processes across the globe as nations look to localise supply lines driven by a shift in industrial policy, diluting industries as market contestability will grow in these friendshored areas. Heavy subsidies will become common in key industries like metal and technology sectors to incentivise localised supply as nations sacrifice growth and efficiency for security in supply.

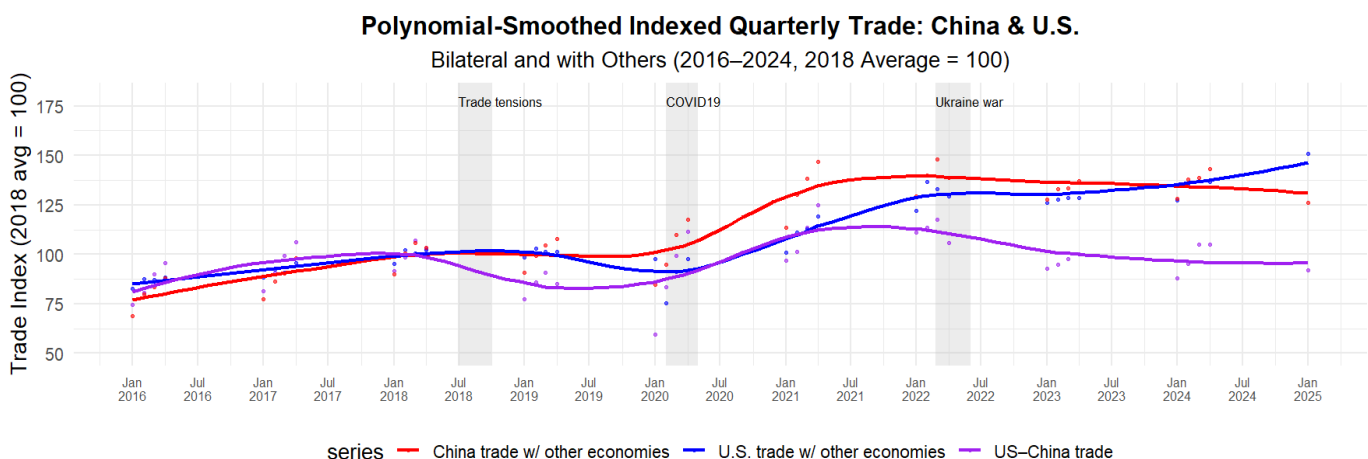


Figure 3

Duplication of production intentionally sacrifices economies of scale and forgone global production is replaced by smaller politic centric markets. This fragmentation creates factor missallocation on two fronts: Firstly, capital expenditure will shift from research towards activating redundant capacity, creating a \$130 billion annual deficit that stalls productivity growth and leaves a stagnant economy (OECD, 2023). This deficit stems from: (1) Idle capacity duplication - maintaining parallel production lines reduces utilization rates and scale economies; (2) R&D diversion- capex shifts from innovation to equipping redundant facilities; (3) Skills mismatch - retraining costs average \$24k/worker (Bain, 2024). And secondly, labour markets experience asymmetric adjustments- high-skilled workers that work in critical friend shored industries capture wage premia at the expense of low-skilled workers who's work is often excluded from GVCs (OCED, 2023) and hence may suffer unemployment. Both these distortions manifest in persistent cost-push inflation and inefficiencies that will regress actual global GDP and GDP growth.

3.0.0 Analytical Appendix:

3.1.0 Analysis

3.1.1 International Trade Analysis: Services vs. Merchandise

In this section we quantify how differences in export-import composition drive cross-country heterogeneity in tariff-shock resilience. It begins by formalising country types, then compares positive and negative trade profiles by using elasticity estimates and counterfactuals to assess the role of merchandise-service linkages and ends with a demonstration of how this fits our Monte Carlo simulation.

Every country trade can be categorised into services and merchandise trade. Furthermore, they can be categorised as:

- 1) A service exporter (service-related exports are greater than service-related imports)
- 2) A service importer (service-related imports are greater than service-related exports)
- 3) A merchandise exporter (merchandise related exports are greater than merchandise related imports)
- 4) A merchandise importer (merchandise related imports are greater than merchandise related exports)

If we combine these dimensions, we obtain the four possible trading profiles. These being a dual exporter (SE-ME); a dual importer (SI-MI) or any mix of the two (SE-MI or SI-ME) that should capture each country's exposure to import-related controls on goods as services trade is not subject to tariffs.

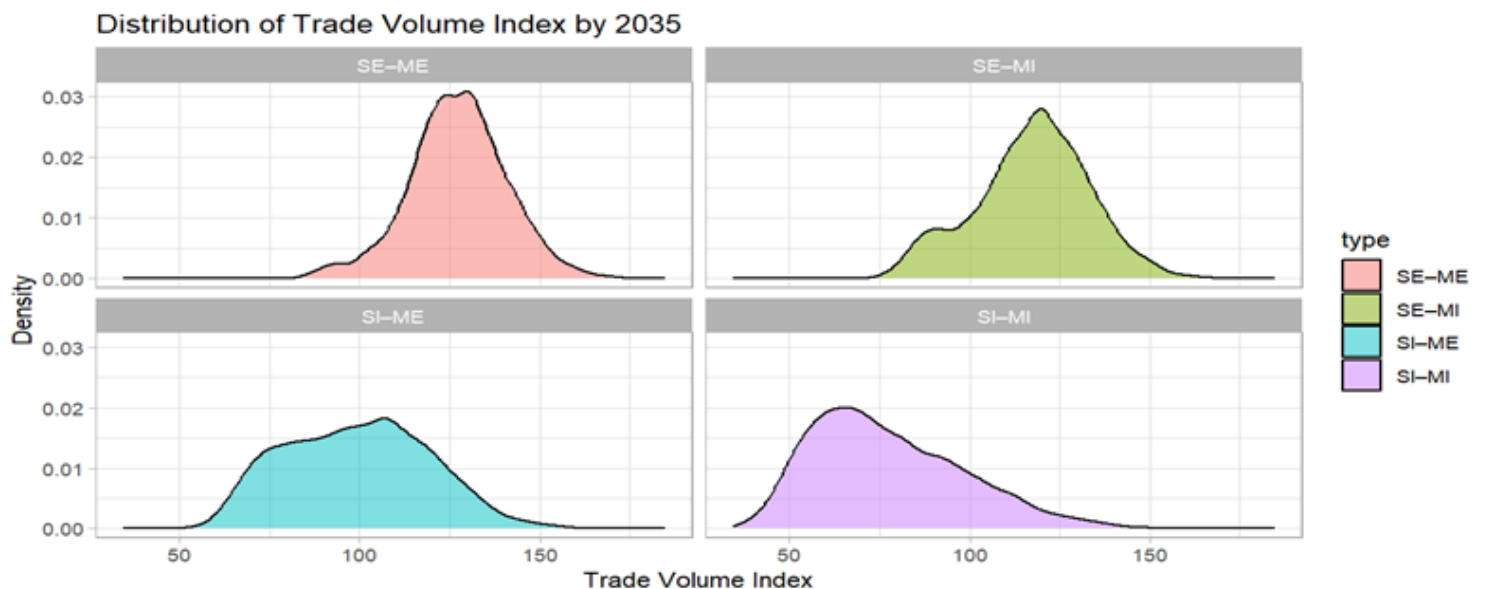


Figure 4

Profile	Country Example	Composition	Resilience
SE-ME	Germany	58% goods, 42% services	High ($\beta = 0.2$)
SE-MI	United Kingdom	45% goods, 55% services	Medium ($\beta = 0.4$)
SI-ME	Vietnam	89% goods, 11% services	Low ($\beta = 0.7$)
SI-MI	Australia	72% goods, 28% services	Critical ($\beta = 0.9$)

The four density plots of 2035 project trade volume quantify how trade structure and structural resilience determine long-run exposure to protectionist shock. SE-MI economies have the highest mean trade volume (127) and narrowest distribution, reflecting their strong service export base and moderate elasticity, which cushions trade shocks. SE-ME countries come second with a mean of 117 and a slightly higher standard deviation, reflecting their balanced export mix and lower aggregate elasticity, underpinned by a structurally independent service sector. By contrast, SI-MI economies- with high dependence on merchandise exports and embedded services face are much more vulnerable, producing a flatter distribution centred around 110 and a longer left tail – due to higher elasticity and lack of diversification. Most vulnerable are the SI-MI countries, with mean trade index below one hundred and significant mass in their 80-100 interval. Their high merchandise dependence, low service trade incidence, high elasticity and resilience mean that a significant proportion of simulations end the forecast with lower trade volumes than in 2025. These distributions illustrate that structural composition, primarily the balance and nature of the service sector in relation to the merchandise sectors, is a fundamental driver of long run trade performance under rising protectionist conditions. Our forecasts for higher protectionism will not impact all nations uniformly but individual, country-based evaluation is required.

3.1.2 Gravity Trade and Elasticities

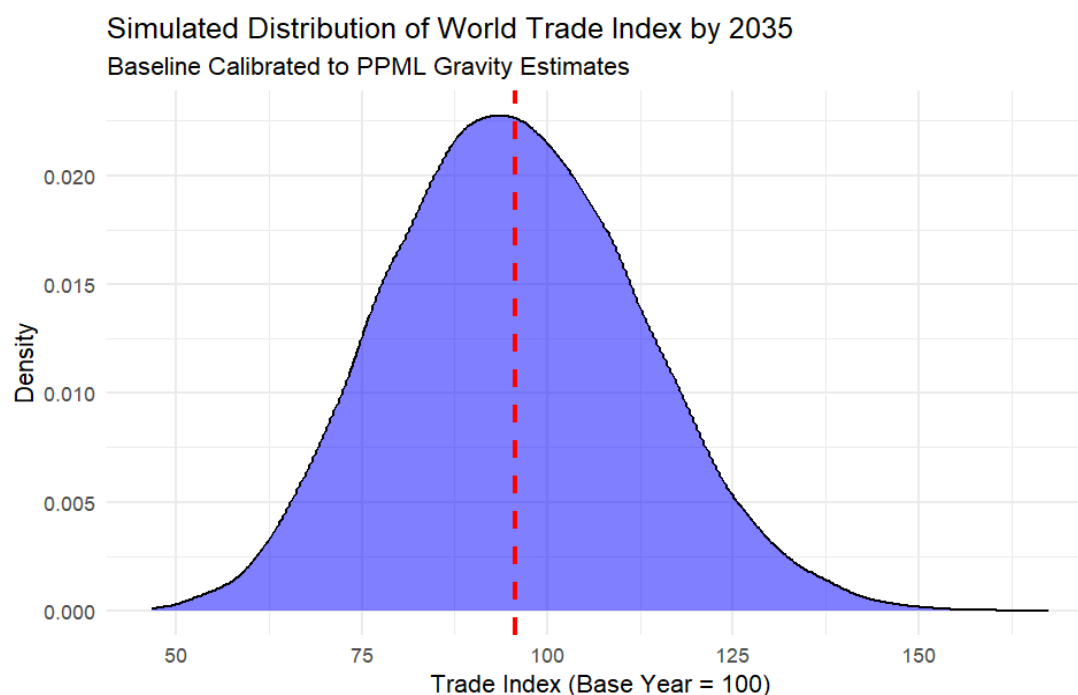


Figure 5

In 2035, the density plot predicts an index level of that related to 2025 levels; but the relatively high deviation around this point is indicative of uncertainty in the global economy. This development suggests that, on average, trade is expected to stay close to current levels; but the high uncertainty shows the existence of a major downside risk. In short, there is moderate chance of major decline in global trade but also moderate upside potential for a period of sustained growth. We learn that the near-unit elasticities for GDP confirm that growth-supporting policies will continue to be the most direct means of encouraging trade, and the distance penalty points to the existence of tangible gains to be made from reducing effective trade costs- but we will come to learn that this is unlikely given the nature of the global geopolitical system.

Part	Term	Statistic
Estimates	GDP (Outgoing)	1.014
Estimates	GDP (Destination)	0.962
Estimates	Population (Outgoing)	-0.499
Estimates	Population (Destination)	-0.352
Estimates	Distance	-0.871
Goodness of fit	R ²	0.918
Goodness of fit	R ² within	0.608

Using our Gravity-Monte-Carlo method, we can begin to understand the economic motivation for friend-and-nearshoring under greater global fragmentation. If, on average, a product traverses through three stages of production, each in a different country, the effective distance penalty is three times our estimated distance elasticity (roughly -2.6). This implies that each positive percentage point change in 'distance' that may influence trade will result in a larger loss in global trade. Friendshoring and onshoring, effectively reduces this distance so that trade routes can continue to function. In our framework, cutting average GVC's from five to three stages would reduce effective elasticity by 1.64 percentage points that would boost intermediate goods trade against a highly fragmented baseline. Secondly, friendshoring often pairs geographical and political proximity. This can further alter the model, should countries wish to align like this; reducing the effective distance parameter further as each border is easier to clear so no further transaction or production costs are incurred. If this harmonization reduces the per-crossing penalty by say 15%, the overall elasticity is $-2.22 (3 \times -0.87 \times 0.85 = -2.22)^1$. This implies that increasingly integrated economies can benefit from outsized trade gains in a friend-shored network.

Beyond trade volume, friendshoring can reduce downside risk. By relocating supply chains to their own shores, or the shores of trusted political allies reduce each nation's exposure to GVC's in which our model shows to be particularly vulnerable during protectionist periods. In other words, reducing interdependence on unpredictable or adversarial trading partners. Production shifts should reflect political allegiances where adjustments to smaller, friendlier supply chains result in shallower and more infrequent protectionist shocks. This aligns with the findings of Handley and Limão (2017), which shows that trade policy uncertainty reduces the volume of trade and delays or reverts entry into foreign markets, particularly where there is uncertainty pertaining to future trade policy. Greater uncertainty and deeper GVC's reinforce each other to enhance the likelihood and magnitude of negative trade shocks therefore friendshoring is a suitable reaction to systemic risk. Therefore, economies that realign their supply chains with dependable partners possess a narrower lower tail in predicted trade flows and therefore are less prone to massive declines, boosting economic resilience in the context of growing fragmentation and policy risk, reflecting their non-exposure.

¹ This is acting under the assumption that each stage in the GVC adds exactly 'one unit' of protectionism, these must all be equal for this interpretation to work. If this is not the case, then it should be a weighted sum of each stage.

Using these conditions and intuition we can use this to make our predictions. We anticipate distance to increase, that would imply a deterioration of bilateral trade in accordance with our elasticity estimates. Simply this deterioration would come in a form of:

$$\frac{\partial \ln(\text{Trade})}{\partial t} = -0.871 \times \frac{\partial \ln(\text{Distance})}{\partial t}$$

Equally,

$$\% \Delta \text{Trade} = -0.871 \times \% \Delta \text{Distance}$$

With the percentage change in distance being strictly positive (protectionism on the rise), indicates the future change in bilateral trade will be strictly negative- that is a contraction in the world economic trade. Earlier we spoke about the positives and negatives of friendshoring, now we understand the positives from an economic standpoint along with the negatives from a welfare and an inflation perspective.

3.1.2 Political Analysis

Recent literature corroborates our prediction that fragmentation will occur along geopolitical lines but does not for our ‘near-shoring’ prediction, rather showing that political proximity is a larger determinant of trade compared to geographical proximity. First, states that operate under similar systems and beliefs have greater ability to negotiate and ratify comprehensive trade agreements or to extend preferential tariff treatments (Mansfield et al., 2000; Sekkel, 2009). Second, pronounced political divergence, increases the likelihood of self-serving economic measures- including sanctions and trade restrictions (Crozet and Hinz, 2020; Hinz and Leromain, 2020). Finally, domestic consumption patterns and preferences can be altered by national sentiment. Citizens increasingly avoid products from politically unaligned nations or increasingly demonstrate enthusiasm for trade with politically aligned nations (Michaels and Zhi, 2010; Heilmann, 2016). Our method takes inspirations from recent research from Jakubik and Ruta (2023) where they are able to measure political discourse by an analysis of voting patterns within the United Nations.

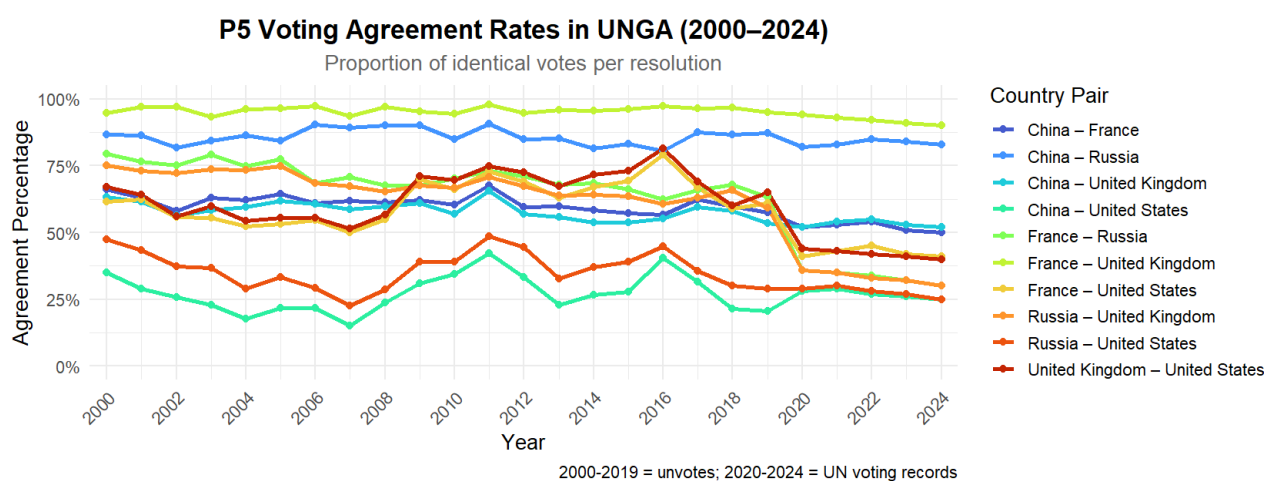


Figure 6

If we abstract the world economy down to the P5 in the UN, which coincides with the major economic players, the evidence suggest that polarisation is increasing. There still remains key alliances in the France-United Kingdom and China-Russia which we can expect to strengthen as a necessary reaction to a weakening in other relations. Recall the earlier graph, where we segmented it by Eastern and Western economies a similar conclusion can be made. European nations are closely allied to other European nations, regardless of EU membership, and they are weakly linked to the US, Japan, and South Korea but Eastern nations are much less aligned with each other. Studies employing gravity models and trade network analysis consistently find that European non-EU nations stay deeply connected to EU markets through arrangements like the EEA and EFTA (Zlati et al., 2021). Integration reflects regionalism, reinforced by aligned regulations, infrastructure and coordination which amplifies trade flows through Europe with noticeable multiplier effects. In contrast, Eastern economies exhibit weaker institutional cohesion focussing on fragmented global value chains rather than coordinated regional demand (EIB, 2025). The shallow integration in Asia translates into less aggregate demand spillovers and less scope for concerted macroeconomic responses. Looking ahead, these contrasting patterns imply that European institutionalized regionalism will bring resilience, innovation benefits, and policy coordination capability in a multipolar global economy, while Asia's fragmented integration will limit its potential to function as a coherent economic bloc (Zlati et al., 2021; European Commission, 2025)

That being said, expect greater European and American integration through the next decade. Three factors drive this: (1) Complementary industrial bases - EU tech + US capital; (2) semi-shared regulatory frameworks (e.g., DPF alignment); (3) Security interdependence:

- (1) The respective economies have developed into strong economic compliments, with the US dominant in the provision of risk capital, extensive capital markets and Europe dominant in innovation value, industrial capabilities and tech R&D. Such complimentary support can lead to greater total factor productivity that generates spillovers into both regions' economic growth and innovation cycles. Both economies have been characterised with a 'productivity puzzle' as of recent and a way to buck this trend is through process and product innovation; through an intimate working relationship, Europe and the US may be able to compete with Eastern productivity and costs.
- (2) Development of semi-shared regulatory frameworks will become a source of deeper integration in order to streamline transatlantic flows. In particular the Data Privacy Framework accounts for the single largest data flow, exceeding that of any other global relationship; decreasing transaction, legal, and administrative costs for firms engaged in transatlantic data flows, leading to cost savings.
- (3) The robust institutions of NATO and collective defence planning, strengthens the US–Europe relationship by assisting in building shared threat perceptions, interoperability, and collective defence investment in emerging military technologies, which becomes ever more crucial as member states grow combined defence expenditures and consolidates industrial capabilities to counter shared geopolitical threats

In the context of global conflict and mistrust which places East vs West on opposing sides: Russia Vs Ukraine, the Middle Eastern conflicts and most recently India Vs Pakistan and Cambodia Vs Thailand. Conflicts are on the rise and will end up further pitting alliances and influences against each other. This should create a self-enforcing loop of conflict that spawns political fallout that spawns conflict creating a world of polarising views and imperialist/nationalist sentiment that prevents economic efficiency making many worse off as a result. We forecast that, under the

current conditions, that this will only increase as ongoing wars see no end in sight with ceasefires being broken in Gaza and Putin playing folly with peace agreements continuing this cycle. Embargos, stipulations and sanctions will endure further harming economic wellbeing of both the sanctioned and those applying the sanction, which will persist into the future.

3.1.3 US-China Dynamics

The decoupling between Chinese and U.S. economies are fracturing global financial markets with much longer-term consequences for global capital allocation, asset prices and cross-border financial interaction. Direct investment into has turned negative (-\$8.3 billion) for the first time in history amid previously mentions regulatory and political differences and uncertainty among other issues with diminishing repatriation flexibility. Portfolio flows had also diminished, particularly from EU and US investors representing a strategic weighting of risk exposure rather than a cyclical withdrawal of funds. In contrast emerging markets more aligned with Western geopolitical influences – Mexico, Vietnam and to a lesser degree India- have experienced significant foreign portfolio investments; most notably in sovereign bonds, equities and export linked manufacturing facilities.

Asset and capital prices will be increasing compartmentalised along geopolitical lines. In U.S equities, domestically focussed sectors- energy, defence and infrastructure- are likely to perform well due to targeted fiscal expenditure targeting that comes part-and-parcel of growing global instability. Similar trends are visible in bond markets: US treasury yields are rising due to high spending and inflationary pressures, while Chinese bond yields are falling as private investment remain weak and a depreciating currency. The Chinese Yuan is expected to fall further relative to the US dollar (around 7.6 Yuan to the Dollar by EOY) as capital continues to leave the country in favour of other nations that are experiencing appreciations – Mexican Peso and Indian Rupee among those currencies strengthening. Changing exchange rates reinforce global divergence, making some exports more competitive while pricing some others out of key markets, further indicate a necessitate for global restructuring in the coming years. Uneven forex adjustment will predicate divergence in the global economy where clear winners and loser are observed. Countries with depreciating currencies, China, benefit from improved competitiveness (partially offsetting tariffs) while emerging markets attract need capital inflows fostering disinflationary growth; by contrast economies facing currency appreciation without productivity gains or capital inflows risk losing export share and deteriorating growth statistics.

Looking forward into out forecasting period, these trends are likely to continue. Funds are likely to opt in favour of raising exposure to decoupling-aligned markets by sacrificing investments into Chinese dedicated markets. Net portfolio inflows to Mexico, India and other aligned periphery economies are likely to increase by 10-15% per annum over the next two years at the expense of sustained outflows from Chinese markets. This in turn could lead to underinvestment, meaning the days of ~8% yearly Chinese GDP growth may be ending. That being said, a partial recovery of Chinese FDI is conceivable but notable domestic reforms will be needed to appeal to an increasingly cautious investor base- this may come in the form of governmental transparency or better geopolitical signalling. Financial markets are also becoming more regionally focussed and capital flows increasingly reflecting geopolitical alignments, this is expected to continue and strengthen as the divide between East and West grows.

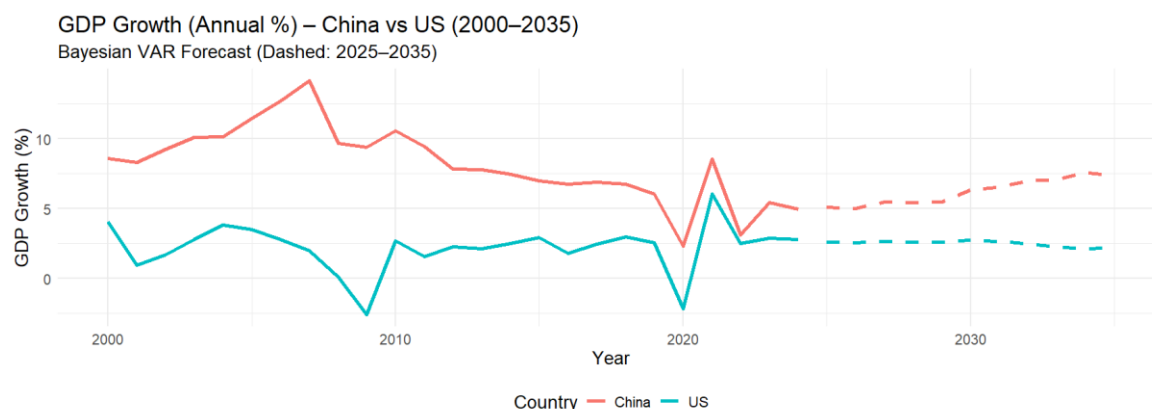


Figure 7

Our forecasts reinforce a growth differential with continuing to converge towards the US economy, with this difference currently reflecting the underlying catch-up effects. Its lower income level allows higher capital returns especially during the transition from heavy industry towards higher value services, technology and domestic consumption markets. This higher growth level will begin to converge to the US economy growth rates beyond the scope of our forecast as China approaches the technological frontier that the US currently operates at. The current US economy is characterised by incremental productivity gains anchoring growth at a reduced level. Advanced economies leave limited scope for growth, resting on major innovation breakthrough or by sustained fiscal expansion that in turn make the US more sensitive to monetary tightening measures and rising debt service burdens.

Under our forecast, China's growth is projected to return to its ~7.5% per year average by 2035. However, our intuition suggests that this may be optimistic. We earlier suggested that foreign capital will begin to flow out of the Chinese economy as part of declining investor confidence and increasing risk; so, this high rate of growth will not be able to be sustained without some alternative method of funding from within the country or from within Asia. Though this graph may be a good indicator of growth in the shorter term (within 5 years) rather than for the longer term (5 years and beyond)

Nonetheless, with the Chinese economy gaining on the American it will begin to serve to be a suitable alternative for the other developing economies. For China, this will reduce the reliance of the West and allow for their own trade in their own currency as the Yuan begins to compete in usage rates to the dollar. For many periphery countries, decisions over whether to align with China or US will take place and risk to further up end the current economic balance; as a result of this we anticipate further posturing to gain political allies in developing Asia, Africa and Latin America, either through aid stipulations or development programs.

3.1.4 Environment and the Green Transition

At the current rate of climate mitigation and adaptation, the cumulative pollution burden will bring economic and systemic instability by the end of our period. One impact is that the productivity of labour is reduced by pollution related disease that drive workforce turnover and absenteeism. An estimated \$3.1 billion lost workdays equal a 0.8% reduction in GDP in high exposure economies in East Asia. Agrarian-dependant areas will face a level of trade reconfiguration as decreasing yield erode supply and hence export competitiveness; China and India can predict \$12-18 billion losses for every year under these conditions (\$180bn loss over the next ten years may cripple the industry). Less polluted regions: Southern America and Australasia will gain agriculture market share. There will be a shift in global food supply; areas that previously

suited a given crop may be unable to grow that crop going forward. That is, cultivar suitability will change meaning that there may exist a period of cross over where a period of poor yield is endured until the nature of crops in that climate is changed – those in the higher and lower latitudes are bound to gain market share in the agrarian sector. Food scarcity will compound inflationary shocks; increasing food prices by 3-5%, eroding household purchasing power that in turn will dampen internal and external consumer demand. So, without mitigation techniques, air pollution and climate changes may impose a lasting cost on future GDP growth and the productivity of human capital.

As with much of our predictions, we can expect this to continue to be the case for the short-term. In the longer-term public policy and private investments will be gauged towards climate mitigation and adaptation sectors that will begin to unwind the economic effects of environmental damage.

There remains an interaction between fragmentation and climate policy. Some nations (US, EU, China) are advocating large scale green subsidies and carbon regulation, these may be uncoordinated and interpreted as protectionist aggression. In turn, this can lead to a “subsidy race” for green technology and can distort comparative advantage and induce trade barriers. Green product trade conditions would ‘cut cross border flows in goods and innovations’ hindering climate tech production and the green transition. Similarly, protectionist green-industry policies risk destabilising global value chains: Europe’s push for near/friend-shoring green production will be resisted by other economies, triggering retaliation. In short, fragmentation impacts the green transition in the same way as any other trade. Fragmentation works against trade efficiency, so it too may impede international public goods like climate mitigation. The IMF and ECB note that in the absence of multilateral coordination, climate action and trade tensions could create a self-enforcing mechanism through the following:

- (1) Initial fragmentation reduces the effectiveness of lone-wolf climate policies and increase costs.
- (2) Higher costs lead to political pressure for protectionist measures to maintain sectoral employment.
- (3) Increased protectionism triggers retaliation
- (4) Reduced cooperation slows technological progress and disincentivises investment.

So, such adaptation to climate impacts is difficult to achieve with global integration and agreements. Hence, we can predict agreements from future COP and SB meetings to be more rigorous and encompassing and urgency increases- as it is in the common interest to do so.

3.1.5 Concluding statements:

Our analysis has provided a forecast rooted in calibrated economic models and logic. They summarise a future global economy more fragmented but explain the cause and consequences of these changes of which advice future policy and investment decisions should. A fragmented economy is an expensive one for the West; global value chains are broken and production relocated to more costly political allies. Growth for all will slow if protectionism continues, higher prices depress consumer demand but higher prices; in food and core index items develop inflationary pressure; further depressing consumer spending power. At the current rate, the future of the global economy looks rather bleak. But the model abstracts from potential innovations in

technology and assumes stable policy environments, as time progresses these models should be recalibrated to continue to forecast based on the most relevant statistical priors. This piece is to not say everything will perform poorly but serves as an indicator of the current trajectory. Of course, there remains a notable probability that policy makers are able to reverse the poor circumstances that drove the reports narrative and continue a regime of liberal free trade, which would greatly benefit the world economy.

4.0.0 Methodology:

4.1.0 Monte Carlo Simulation of Global Trade Under Rising Protectionism

We use indexed values for a measure of trade volume such that a recursive growth identity where trade volume in the next period is equal to the product of a shock multiplier and the index of the previous period.

Each simulation year has a probability of experiencing a protectionist shock, this initial shock probability is type specific and reflects the structural exposure to trading frictions – this is specifically related to the merchandise trade identifier. Following each shock year, the probability of incurring another shock year increases, up to a probability of one. We model it in this way as it mimics the dynamics of trade wars and retaliation that are common under a scenario with a weakened WTO. This mechanism allows for the simulation to capture cumulative and compounding risk over the decade.

To provide a robust formula to determine the shock multiplier, which is as follows.

$$\text{Shock multiplier} = \text{Elasticity} \times (\text{Share}_{\text{merchandise}} + \text{Resilience}_{\text{services}} \times \text{Share}_{\text{services}})$$

Where elasticity measures trade volume responsiveness to tariff shocks, $\text{Share}_{\text{merchandise/services}}$ measures the proportion of each country type's total trade volume is made up of merchandise and services (by default these should sum to one) and $\text{Resilience}_{\text{services}}$ is a parameter between [0,1] that reflects the characteristics and independence of the services sector where a lower value represents higher independence.

Together this formulation allows for shocks to vary in both frequency and intensity, in a manner that is tailored to underlying trade structures of each country.

In the model we randomise all variables not limited to, shock impact, service resilience and trade share. This is as the country types we defined earlier are aggregates of multiple, diverse economies that happen to fall in the same category. That is saying Germany is equal to Singapore which is obviously not true. Fixed parameters imply all within the same group have the identical trade structures, which is unrealistic. By randomising each parse, the outcomes reflect within-group variation that are critical for informing decisions. Also, fixed parameters ignore uncertainty as so now; by incorporating uncertainty over future trade structures, our model gives a more robust view of future world trade.

4.2.0 ARIMA Forecasting:

An ARIMA forecast is comprised of three components that, working together, can strike a balance between accuracy and interpretability for long run time series data. Our data set begins in 1960 and runs until Q4 2024 featuring yearly observations of GDP, a variable suited to ARIMA with its non-stationary yet autoregressive properties. These three components are: (1) an autoregressive part, that models output as a linear function of past values where coefficients of output imply that recovery from shocks occurs incrementally; (2) an integrated part that transform trending data into a stable form that allows it to be forecasted; and (3) a moving-average part which captures short terms irregularities that the autoregressive formula may not be able to isolate.

Should you be interested an ARIMA (p, d, q) model takes the form:

$$\nabla^d Y_t = \phi_1 \nabla^d Y_{t-1} + \dots + \phi_p \nabla^d Y_{t-p} + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \dots + \theta_q \varepsilon_{t-q}$$

Where:

Y_t is the GDP in time t.

$\nabla^d Y_t$ is Dth difference of the time series at time t. That is how much the value has changed compared to previous values – this comes from the integrated part.

ϕ_1 represents the autoregressive coefficient, which is how much the current differenced value from $\nabla^d Y_t$ is dependant on the previous value.

ε_t is the error term at time t. This captures unexpected variation from the predicted value and the actual value from 1960-2025.

θ_1 is the moving average coefficient at time t. For example, if θ_1 is our moving average then today's average is influenced by a shock that occurred 1 time set ago.

The model is implemented as per the Box-Jenkins approach (Box et al., 2015), model order being established based on information criteria (AIC) and diagnostic checks that ensure residuals converge towards maximum accuracy. The code does this by iteratively checking each ARIMA (X, Y, Z) combination until error stats are minimised by comparing prior known values against the created prediction function. The minimised function will then be used for the forecasting.

4.3.0 BVAR Forecasting:

A Bayesian Vector Autoregression is a method of modelling times series data between multiple related variables simultaneously where their direct effects on each other can be estimated from the data. In this method, each variable is expressed as a function of their own lags, and as a function of the lags of all other relevant variables with Bayesian priors weighting model parameters. These priors are based on economic theory or expert opinion, and serve to regularize the estimates, shrinking them toward more reasonable or parsimonious values. This is particularly crucial in macroeconomic forecasting, where the number of parameters may be relatively large compared to the available sample size: the Bayesian approach prevents overfitting and stabilizes forecasts when data are scarce. Independent distributions are attributed to each parameter and are estimated by running a Monte Carlo simulation. Predictive distributions for future GDP outcomes are generated by simulating from the model, yielding both point forecasts and credible intervals that reflect full parameter and process uncertainty.

BVAR naturally fits the spillovers and feedback loops of large, interconnected economies. ARIMA can only model each country's GDP separately, but the BVAR can capture how shocks or

trends in one economy directly affect the other systemically, producing more realistic and policy-relevant forecasts. In addition, the Bayesian framework enables the incorporation of reasonable prior beliefs about persistence or volatility meaning research and practical experience confirm that BVAR tends to outperform univariate models like ARIMA- in several circumstances- in macro-financial forecasting, especially when countries or variables move together or respond to common shocks.

	Country	ME	RMSE	MAE	MAPE	MASE
1	China	-0.02044280	5.368085	3.497699	148.8042	0.6750588
2	US	0.01573611	2.405996	1.741469	162.8797	0.5757935

Figure 8: BVAR accuracy statistics.

	Country	ME	RMSE	MAE	MAPE	MASE
1	China	0.007085997	5.094043	3.343073	56.76562	0.8956735
2	United States	0.309140196	2.062956	1.449147	235.31274	0.7712862

Figure 9: ARIMA accuracy statistics.

Convention places MASE (Mean Absolute Scaled Error) as the deciding factor when determining method accuracy. It is a scale-independent metric that compares the absolute error of the method to the average error of the naïve method: a score greater than 1 indicates that the method performs worse than the naïve method meaning scores approaching zero indicate strength above the naïve method. Given MASE is lower in BVAR compared to ARIMA, BVAR should be used for our forecasts.

4.4.0 Gravity model

First, we estimate a standard gravity bilateral trade following a PPML (Poisson Pseudo-Maximum Likelihood) framework to determine the joint effects of economic mass, population and ‘trading distance’, which is a composite measure that can include geographical distance, cultural distance and the existence of trade barriers. Second, we employ the coefficients to construct a synthetic world trade index and then project its evolution in the coming decade based on a Monte Carlo simulation that replicates stochastic shock and growth events. The data set we use comprises of 490,000 origin-destination pairs from the CEPII Gravity database (Conte. M et al, 2022). For each pair we observe, GDP, Population, geographical distance among other measures in US dollars and in purchasing power parity, cleansing the data set for any missing or zero trade values. Then we take the natural logarithm for all continuous variables so that the coefficients estimated have elasticity interpretations.

We formulate:

$$E[\text{trade flow}_{ij}] = \exp(\beta_1 \ln GDP_i + \beta_2 \ln GDP_j + \beta_3 \ln Pop_i + \beta_4 \ln Pop_j + \beta_5 \ln Dist_{ij} + \alpha_i + \alpha_j)$$

where $\alpha_i + \alpha_j$ are export and importer fixed effects, respectively, and, by default, incorporate standard errors within.

The approximation yields GDP elasticity close to unity for exporter and importer alike, that is a one percent increase in GDP for either country will result in an equal gain in bilateral trade. The result confirms the central gravity-model prediction that economic mass produces trade in proportionally similar fashion. In contrast, population effects arrive negatively but are statistically insignificant from zero after controlling for GDP, so that aggregate output rather than headcount determines trade potential. Distance performs with a strong negative relation, with elasticity -0.87 where a 1% increase in distance reduces trade by 0.87. The models overall R^2 of .918 indicates good explanatory power and a within control of R^2 of .607 confirms that that covariates themselves can be used to explain 60% of trade variation, leaving 40% left to unobserved variables – which is acceptable with respects to global trade where there could be hundreds of influencing variables (academic benchmark lies between 50-70%).

With these parameter estimates, we initialize each simulation path at the PPML-implied trade level. We simulate 10,000 paths over the ten-year forecast, where the previous year is multiplied by the augmenting variable $\exp(\delta_t)$ of which is taken from either a shock distribution or a growth distribution. This is as follows:

$$T_{t+1} = T_t \times \exp(\delta_t), \quad \delta_t \begin{cases} N \sim (-0.10, 0.05^2) \\ N \sim (0.03, 0.02^2) \end{cases}$$

This method will formulate an index number for 2035, from which we can determine whether global trade will increase, decrease or remain stagnant in the coming year under our parameters and assumptions. Plotting these index numbers onto a density plot should allow us to approximate a normal distribution to conduct probability analysis.

4.5.0 Political modelling

Modelling political discourse is particularly troublesome as there is no true measure for fallout, so we employ a series of proxy-measures to best gauge distance. The UN records is one way to do this, but it is not without its limitations. The database the UN provides is part of these limitations, not only does it include the major resolutions but includes votes over meeting times and trivial matters that may skew results by indicating political agreement despite not having any real impact on the real world. Our method works through and removes these trivial votes so that only the vote remain that provide the most insight into the political sphere.

From here it is rather simple we work through on a vote-by-vote basis matching agreements: yes, to yes, no to no and abstain to abstain. By doing this for each pair of nation we calculate an agreement proportion where a score closer to one represents more aligned nations. With this we graphed a static- as of Dec 2024- and a time series plot of the P5 nations.

5.0.0 Code Appendix:

For formatting reasons, as to not extend the size of the document any further, the code appendix will follow as a series of links to a [GitHub repository](#). The links contain both the code for the method and the code for the graphs and tables included in the report.

Figure 1:

https://github.com/JoeHardingCODE/10-Year-Economic-Forecast/blob/Code/CODE_FRAMEWORK_DIAGRAM.R

Figure 2:

https://github.com/JoeHardingCODE/10-Year-Economic-Forecast/blob/Code/CODE_VOTING_ALIGNMENT.R

Figure 3:

https://github.com/JoeHardingCODE/10-Year-Economic-Forecast/blob/Code/CODE_CHINA_US_TRADE.R

Figure 4:

https://github.com/JoeHardingCODE/10-Year-Economic-Forecast/blob/Code/CODE_MC_PROFILE.R

Figure 5:

https://github.com/JoeHardingCODE/10-Year-Economic-Forecast/blob/Code/CODE_GRAVITY_MC.R

Figure 6:

https://github.com/JoeHardingCODE/10-Year-Economic-Forecast/blob/Code/CODE_VOTING_ALIGNMENT_TIME.R

Figure 7:

https://github.com/JoeHardingCODE/10-Year-Economic-Forecast/blob/Code/CODE_CHINA_US_GROWTH_BVAR.R

but takes information from the following:

https://github.com/JoeHardingCODE/10-Year-Economic-Forecast/blob/Code/CODE_CHINA_US.R

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Thank you for reading.
Joe Harding (UG)